EVICTGuard and RISKSTOP: A Budget-Aware Framework for Evidence-Preserving LLM-Based SOC Investigation

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Security Operations Centre (SOC)

- A Security Operations Centre (SOC) is a centralised security environment that continuously monitors an organisation's systems for cyber threats.
- It collects alerts and telemetry from networks, endpoints, cloud services, applications, and identity systems, which analysts investigate to detect and understand threats.
- Large volumes of alerts, false positives, duplicate events, and scattered evidence can make SOC investigations difficult and time-consuming.

LLM-Based SOC Investigation

- **LLMs assist SOC analysts in security investigation**
  - Generate queries, retrieve and analyse telemetry, and support alert triage.
- **Investigations operate under limited resources**
  - Limited context and investigation budgets restrict how much evidence can be processed at a time.
- **Two key investigation challenges**
  - **Evidence loss:** Critical evidence may be evicted by excessive telemetry.
  - **Premature stopping:** Early stopping may leave important evidence undiscovered.

Literature Survey : Evidence Reduction & Alert Prioritisation

Author/year	Methodology	Advantages	Limitations
Hassan et al., 2019 (NoDoze)	Uses provenance graphs and historical context to identify relevant dependency paths and reduce irrelevant alert information.	Reduces false alarms while retaining important causal information.	Does not consider bounded LLM context, query budgets, or adversarial telemetry flooding.
Wang et al., 2024 (AlertPro)	Combines alert properties, sequence context and historical feedback to prioritise alerts.	Helps analysts identify high-priority alerts in multi-step attacks.	Focuses on alert prioritisation rather than evidence management during an ongoing investigation.
Xiao et al., 2025 (GRAIN)	Reduces redundant alerts, constructs causal attack graphs, estimates causal importance, and selects important attack paths.	Preserves causally significant events and reconstructs multi-step attacks.	Does not control sequential LLM queries or determine when an investigation should stop.

Literature Survey : LLM-Based SOC Investigation

Author/year	Methodology	Advantages	Limitations
Wei et al., 2025 (CORTEX)	Multi-agent LLM system for behaviour analysis, evidence retrieval and security reasoning.	Evidence-grounded and auditable SOC investigation.	No evidence preservation under telemetry overload; higher cost and latency.
Ndichu et al., 2026 (PACT)	Active learning with drift detection and analyst feedback for SOC alerts.	Reduces false-positive burden and analyst workload.	Focuses on alert labelling, not multi-step investigation.
Chona et al., 2026 (CDB)	LLM agents hunt through large Windows logs using SQL under a query budget.	Tests open-ended threat hunting with ground truth and resource limits.	Benchmark only; no evidence-protection mechanism.

Literature Survey : Budget & Decision Control

Author/year	Methodology	Advantages	Limitations
BAGEN (Lin et al., 2026)	Estimates lower/upper bounds of remaining resources during agent execution and identifies when successful completion is unlikely.	Improves budget awareness and can reduce unnecessary resource consumption through early stopping.	General-purpose agent setting; does not consider security evidence coverage or risk of missing attack-critical evidence.
Dual-Threshold Conformal Deferral (Şahin & Mert, 2026)	Uses separate thresholds for automatic escalation, automatic closure, and analyst deferral.	Provides controlled automation with explicit error budgets and human deferral.	Focuses on terminal alert-level decisions rather than sequential evidence acquisition and investigation stopping.

Literature Survey : LLM Alert Classification & SOC Challenges

Author/year	Methodology	Advantages	Limitations
Tariq et al., 2025	Systematic survey of alert-fatigue causes, mitigation approaches, automation, augmentation, and human–AI collaboration in SOCs.	Identifies workload, false alarms, capacity limitations, and human deferral as important SOC challenges.	Broad survey; does not provide a query-level evidence-preservation or investigation-stopping mechanism.
Rieger et al., 2026	Evaluates multiple LLMs for security-alert classification and priority assignment and compares them with traditional ML classifiers.	Shows strong potential for LLM-assisted alert classification and highlights accuracy–cost trade-offs.	Focuses on alert-level classification and prioritisation rather than multi-step evidence acquisition and preservation.

Outcome of Literature Survey

- Existing studies address:
  - Alert reduction and prioritisation
  - Evidence correlation and LLM-based investigation
  - Budget-aware investigation and human deferral
- These capabilities are largely addressed as separate problems.
- **Evidence Gap:** Existing methods do not specifically protect attack-critical evidence from eviction during telemetry flooding.
- **Decision Gap:** Existing methods do not jointly control sequential investigation continuation or stopping based on evidence sufficiency, uncertainty, and remaining budget.

Problem Statement and Objectives of the Project

To develop a budget-aware SOC investigation framework that protects attack-critical evidence from telemetry flooding and prevents premature investigation termination under incomplete and noisy telemetry.

Objectives

1. Develop **EVICTGuard** to protect attack-critical evidence from loss or eviction during telemetry flooding.
2. Develop **RISKSTOP** to control sequential investigation using evidence sufficiency, progress, and remaining budget.
3. Improve investigation reliability under incomplete, noisy, and duplicated telemetry.
4. Evaluate the framework under varying telemetry flooding, noise, missing evidence, and resource constraints.

Methodology To Be Used : Datasets

- **Cyber Defense Benchmark (CDB) – Chona et al., 2026 [7]** Contains agentic threat-hunting scenarios over large Windows logs, with query-limited investigation episodes and ground-truth labels; used as the primary dataset for evaluation.
- **OTRF Security-Datasets – Rodriguez, 2020 [11]** Contains reproducible, labelled attack-scenario logs collected from simulated adversary techniques; used to construct controlled investigation scenarios.
- **Noise Injection:** Telemetry flooding, duplicated events, benign-looking floods, and adversarial mimic-flooding are synthetically injected into both datasets to evaluate evidence preservation and stopping behaviour.

Methodology To Be Used : Overall Framework

- **Telemetry & Query Layer:** Prepare telemetry and control data retrieval and query cost.
- **Evidence Ledger:** Store retrieved evidence with source and integrity information.
- **Evidence–Context Separation:** Preserve raw evidence while maintaining bounded LLM context.
- **LLM Investigator:** Analyse evidence, form hypotheses, and generate queries.
- **RISKSTOP Controller:** Decide whether to continue or stop the investigation.

Methodology To Be Used : Overall Framework

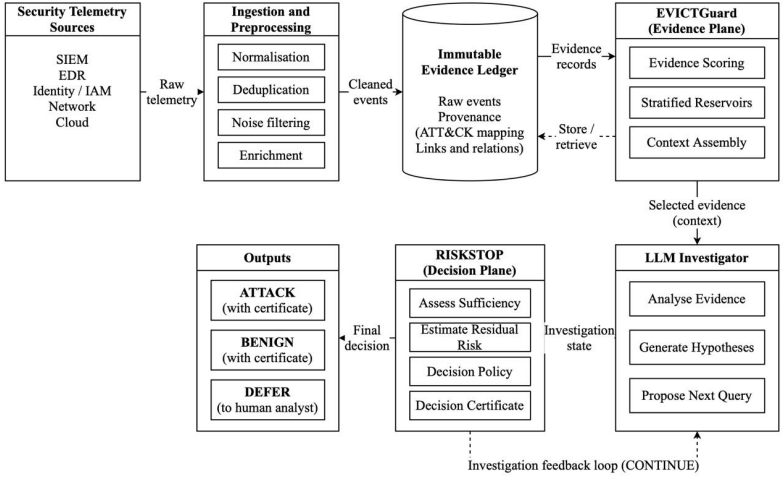


Fig. 1: Overall architecture of the proposed framework.

Methodology To Be Used : EVICTGuard

- **Evidence-Value Estimation:** Scores evidence based on relevance, sensitivity, corroboration, attack-stage importance, and redundancy.
- **Stratified Retention:** Preserves diverse and important evidence across sources, entities, attack stages, and contradictions.
- **Duplicate Reduction:** Groups redundant events to reduce unnecessary context consumption while retaining representative evidence.
- **Protected Evidence:** Reserves context space for high-impact and independently corroborated evidence.

Methodology To Be Used : EVICTGuard

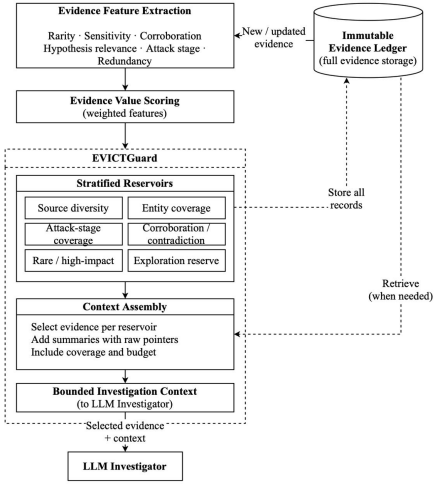


Fig. 2: EVICTGuard evidence-management flow.

Methodology To Be Used : RISKSTOP

- **State Evaluation:** Tracks evidence coverage, unresolved hypotheses, missing telemetry, and remaining budget.
- **Evidence Risk:** Estimates the risk of important malicious evidence remaining undiscovered.
- **Query Value:** Determines whether further evidence acquisition is worthwhile within the remaining budget.
- **Decision:** Selects CONTINUE, ATTACK, BENIGN, or DEFER based on evidence sufficiency and uncertainty.
- **Decision Certificate:** Records the verdict, supporting evidence, unresolved issues, and resources consumed.

Methodology To Be Used : RISKSTOP

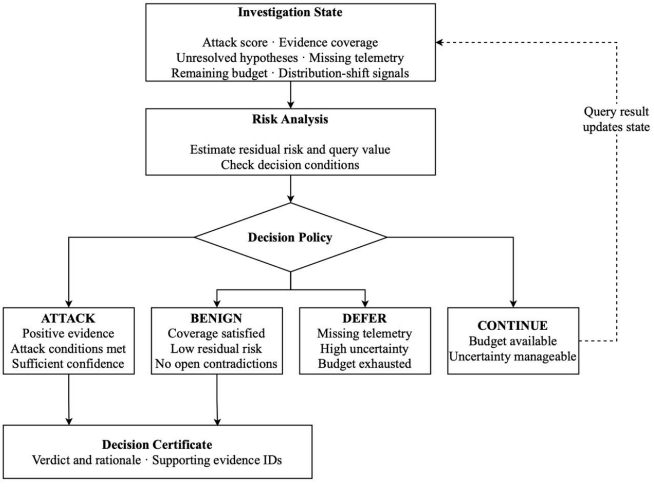


Fig. 3: RISKSTOP investigation-control flow

References

[1] Wajih Ul Hassan, Shengjian Guo, Ding Li, Zhengzhang Chen, Kangkook Jee, Zhichun Li, and Adam Bates. “NoDoze: Combatting threat alert fatigue with automated provenance triage.” In *Proceedings of the Network and Distributed System Security Symposium (NDSS)*, 2019.

[2] Xiang Wang, Xiaochun Yang, Xinyu Liang, Xueqiang Zhang, Wei Zhang, and Xusheng Gong. “Combating alert fatigue with AlertPro: Context-aware alert prioritization using reinforcement learning for multi-step attack detection.” *Computers & Security*, 137:103583, 2024.

[3] Fengrui Xiao, Shuangwu Chen, Jian Yang, Huasen He, Xiaofeng Jiang, Xiaobin Tan, and Dong Jin. “GRAIN: Graph neural network and reinforcement learning aided causality discovery for multi-step attack scenario reconstruction.” *Computers & Security*, 148:104180, 2025.

[4] Bowen Wei, Yuan Shen Tay, Howard Liu, Jinhao Pan, Kun Luo, Ziwei Zhu, and Chris Jordan. “CORTEX: Collaborative LLM agents for high-stakes alert triage,” 2025.

[5] Shahroz Tariq, Mohan Baruwal Chhetri, Surya Nepal, and Cecile Paris. “Alert fatigue in security operations centres: Research challenges and opportunities.” *ACM Computing Surveys*, 57:1–38, 2025.

[6] Samuel Ndichu, Tao Ban, Seiichi Ozawa, Takeshi Takahashi, and Daisuke Inoue. “PACT: Reducing alert fatigue in low-prevalence SOC streams with triggered active learning,” 2026.

References

[7] Alankrit Chona, Igor Kozlov, and Ambuj Kumar. “Cyber defense benchmark: Agentic threat hunting evaluation for LLMs in SecOps.” Technical Report, Simbian AI, April 2026.

[8] Yuxiang Lin, Zihan Wang, Mengyang Liu, Yuxuan Shan, Longju Bai, Junyao Zhang, Xing Jin, Boshan Chen, Jinyan Su, Xingyao Wang, Jiaxin Pei, and Manling Li. “BAGEN: Are LLM agents budget-aware?,” 2026.

[9] Fatih Şahin and Necibe Sare Mert. “Dual-threshold conformal deferral for trustworthy security alert triage.” *Electronics*, 15(18):4084, 2026.

[10] Matthias Rieger, Atif Shah, Abu Alam, and Md. Jakir Hossain. “Possibilities and limitations of using large language models (LLMs) for alert classification and prioritisation in security operations centers (SOCs).” *Expert Systems with Applications*, 331:133194, 2026.

[11] Roberto Rodriguez. “Security-Datasets: Re-play security events,” GitHub repository, 2020.

Timeline of the Project

Phase	Activities	Status
Phase 1	Problem identification, domain understanding, literature survey	Completed
Phase 2	Problem formulation, objectives	Completed
Phase 3	Design and implementation of **EVICTGuard**	Ongoing
Phase 4	Design and implementation of **RISKSTOP**	Upcoming
Phase 5	Experimental setup, evaluation, analysis, and report completion	Upcoming